

Full Length Article

An experimental and modeling study on the oxidation of ammonia-methanol mixtures in a jet stirred reactor

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ABSTRACT

Methanol and liquid ammonia can form a mixture with better combustion quality. In order to investigate the ammonia (NH₃)/methanol (CH₃OH) mixture oxidation process, jet-stirred reactor (JSR) oxidation experiments were conducted at atmospheric pressure and temperature of 700–1200 K. The concentration of NH₃ in the NH₃/CH₃OH mixture was 80%, 60%, and 40%, respectively, for equivalent ratios of 0.8, 1.0, and 1.2. By using synchronous vacuum ultraviolet photoionization mass spectrometry, important species (such as CH₂O, CO, NO, NO₂, N₂O) involved in the oxidation reaction were identified and quantified. The original chemical kinetic model was validated and optimized based on experimental data, which improved the model's ability to predict NH₃/CH₃OH oxidation process (the overall improvement is approximately 81.3% compared to the original mechanism). The results demonstrated that methanol accelerated the rate of ammonia oxidation, and methanol was rapidly oxidized at high methanol content. The sensitivity analysis of reaction pathways with varying equivalent ratios revealed that HO₂ generated a substantial amount of OH radicals and facilitated the oxidation of methanol and ammonia. During the oxidation of ammonia, nearly half of the NO produced NO₂ through NO + HO₂ = NO₂ + OH, and about 60% of the NO came from the reduction reaction of NO₂. More than 80% of the N₂O was produced by NH₂ + NO₂ = N₂O + H₂O, and most of the N₂O was converted to N₂, while NH₂ promoted the conversion of NO_x to N₂.

1. Introduction

In recent years, ammonia (NH₃) has garnered considerable interest as a carbon-free, environmentally friendly fuel. The fertilizer and chemical industries make extensive use of ammonia, which enabled highly developed generation, transportation, and storage technologies. Additionally, through the Haber-Bosch process, ammonia can be produced using renewable energy, making it a new type of green fuel [1]. Ammonia is an effective hydrogen carrier due to its high hydrogen content per unit volume and significantly higher volumetric energy density than hydrogen [2,3]. However, there are limitations to the direct use of ammonia as a fuel in power machineries. Firstly, ammonia has a low reactivity, a high autoignition temperature, a minimum ignition energy that is more than 60 times that of gasoline, a laminar flame speed that is only about one-sixth that of gasoline, and a narrow but high maximum combustible range, making it difficult to apply in internal combustion engines and gas turbines [4]. Secondly, the combustion of ammonia results in increased nitrogen oxide emissions (NO, NO₂, N₂O,

etc. Included among these, NO as an odorless, colorless, and toxic gas, can pollute water, soil, and the atmosphere. Inhalation of NO can damage the respiratory system. NO₂ is a toxic, irritating, reddish-brown gas that, when exposed to sunlight, can trigger a chain reaction with hydrocarbons to form photochemical smog. It typically poses risks to human health, including irritation of the respiratory mucosa, lung damage, and neurological disorders. N₂O is an odorless, sweet-tasting greenhouse gas that can contribute to ozone layer depletion and ozone hole formation. Depending on the conditions of ammonia combustion, nitrogen oxides can change due to the oxidation of nitrogen and incomplete combustion of ammonia [5]. To optimize the ammonia combustion, ammonia is usually mixed with other fuels that have superior combustion characteristics. Hydrogen, methane, DME are commonly used as co-mixer. Using the heat flux method, Han et al. [6–8] measured the laminar burning velocities of NH₃/H₂, CO, CH₄, N₂O, and syngas mixtures at various equivalence ratios and NH₃ concentrations at ambient temperature and pressure. It was found that the NH₃/H₂ mixture is proved to be the most effective method for increasing the combustion rate of NH₃ mixtures with the highest laminar burning

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Nomenclature

JSR	Jet-stirred reactor
DME	Dimethyl ether
SCR	Selective catalytic reduction
EHN	2-ethylhexyl nitrate
PODE2	Polyoxymethylene dimethyl ether
ATAC	Active thermal atmosphere combustion
φ	Fuel-air equivalence ratio
T	Temperature, [K]
HPST	High-pressure shock tube
IDT	Ignition delay time
RMG	Reaction mechanism generator
SVUV	Synchrotron vacuum ultraviolet, [mol]
NSRL	National Synchrotron Radiation Laboratory
GC	Gas chromatography
PSR	Perfectly stirred reactor
A	Refers to prefactor
n	Temperature factor
Ea	Experimental activation energy, [J·mol ⁻¹]
ROP	Rate of production

through the effect on the concentration of H and OH radicals. A reduction in ammonia gas concentration may slow the overall reaction rate, resulting in a reduction in flame velocity. Shrestha et al. [10] measured the laminar burning velocities of NH_3/H_2 mixtures at equivalence ratios ranging from 0.8 to 1.4 and pressures ranging from 1 to 10 bar at a temperature of 473 K using the constant volume combustion bomb. Increasing the initial temperature, hydrogen content and, oxidizer oxygen content could increase the laminar flame speed, while increasing the initial pressure decreased it. Similarly, Yin et al. [11] measured the laminar flame speed of $\text{NH}_3/\text{DME}/\text{air}$ mixtures with different equivalence ratios under high pressure and high temperature conditions, using the constant volume combustion bomb. The experimental data revealed that the laminar flame speed increased with the temperature and pressure. It peaked at an equivalence ratio of approximately 1.1. Consequently, it is evident that the combined method of blending ammonia with other fuels and improving the initial combustion conditions can effectively increase the ammonia flame velocity. Improvements can also be made to suppress NO_x emissions by controlling combustion conditions and blending techniques. The research of Kurata and Okafor demonstrates that two-stage combustion reduces nitrogen oxide emissions and improves ammonia combustion efficiency [12,13]. Experiments conducted by Li et al. [14] on the combustion characteristics of H_2 and NH_3 at various equivalence ratios and the formation of NO_x revealed that mixed ammonia with fuels can significantly reduce combustion

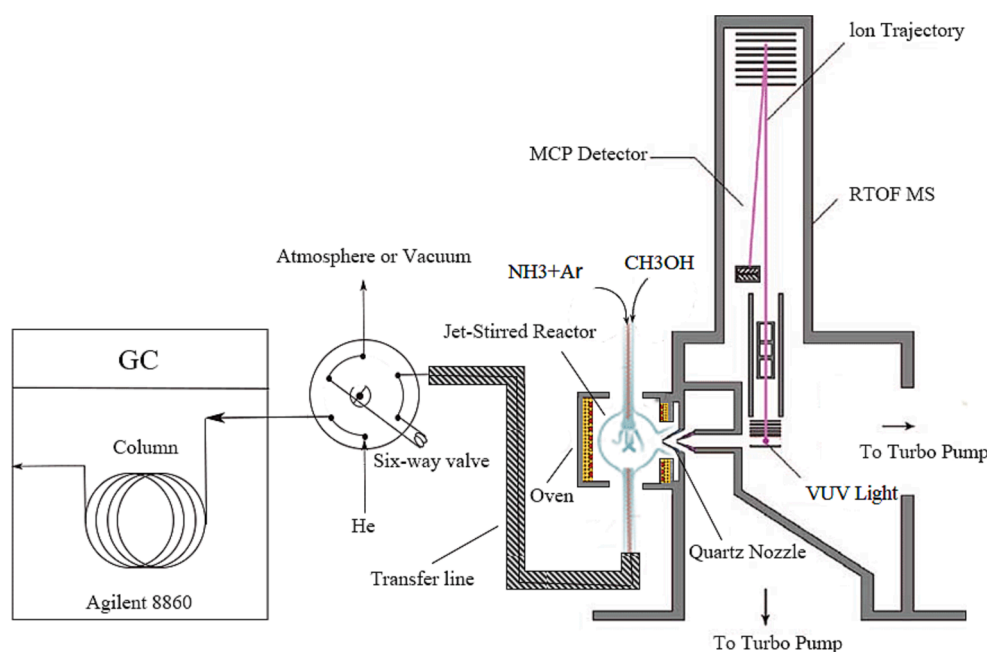


Fig. 1. The JSR system equipped with SVUV-PIMS and GC/GC-MS.

Table 1

Molar fraction of each fuel.

Group	NH_3	CH_3OH	O_2	Ar	$\text{NH}_3/\text{CH}_3\text{OH}$	φ
1	0.008	0.012	0.024	0.956	2:3	1.0
2	0.012	0.008	0.021	0.959	3:2	1.0
3	0.016	0.004	0.0225	0.9575	4:1	0.8
4	0.016	0.004	0.018	0.962	4:1	1.0
5	0.016	0.004	0.015	0.965	4:1	1.2

velocity. Nevertheless, the NH_3/CH_4 mixture had a smaller impact on improving the combustion rate with moderate enhancement of flame burning velocity. Okafor [9] discovered that the effect of ammonia chemistry on the burning rate of methane/ammonia flames is mainly

temperature and NO_x emissions. Especially when oxygen concentration is high, the relative reduction in NO_x emissions is more pronounced. Wang et al. [15] investigated the ignition characteristics of NH_3/H_2 mixtures under varying boundary conditions and discovered that the formation of NO_x was essentially unaffected by the hydrogen addition ratio. The trend of NO_x formation was weakened as the equivalence ratio increased, and the NO_x emission was nearly eliminated when the equivalence ratio was greater than 1.6. Sun et al. [16] conducted a simulation-based study to investigate the performance of selective catalytic reduction (SCR) aftertreatment technology in reducing NO_x and ammonia emissions in an ammonia-diesel dual-fuel engine. This research provides theoretical guidance for reducing NO_x emissions in ammonia and its blended fuel engines. Generally, it was evident that combining NH_3 with other fuels reduces NO_x emissions.

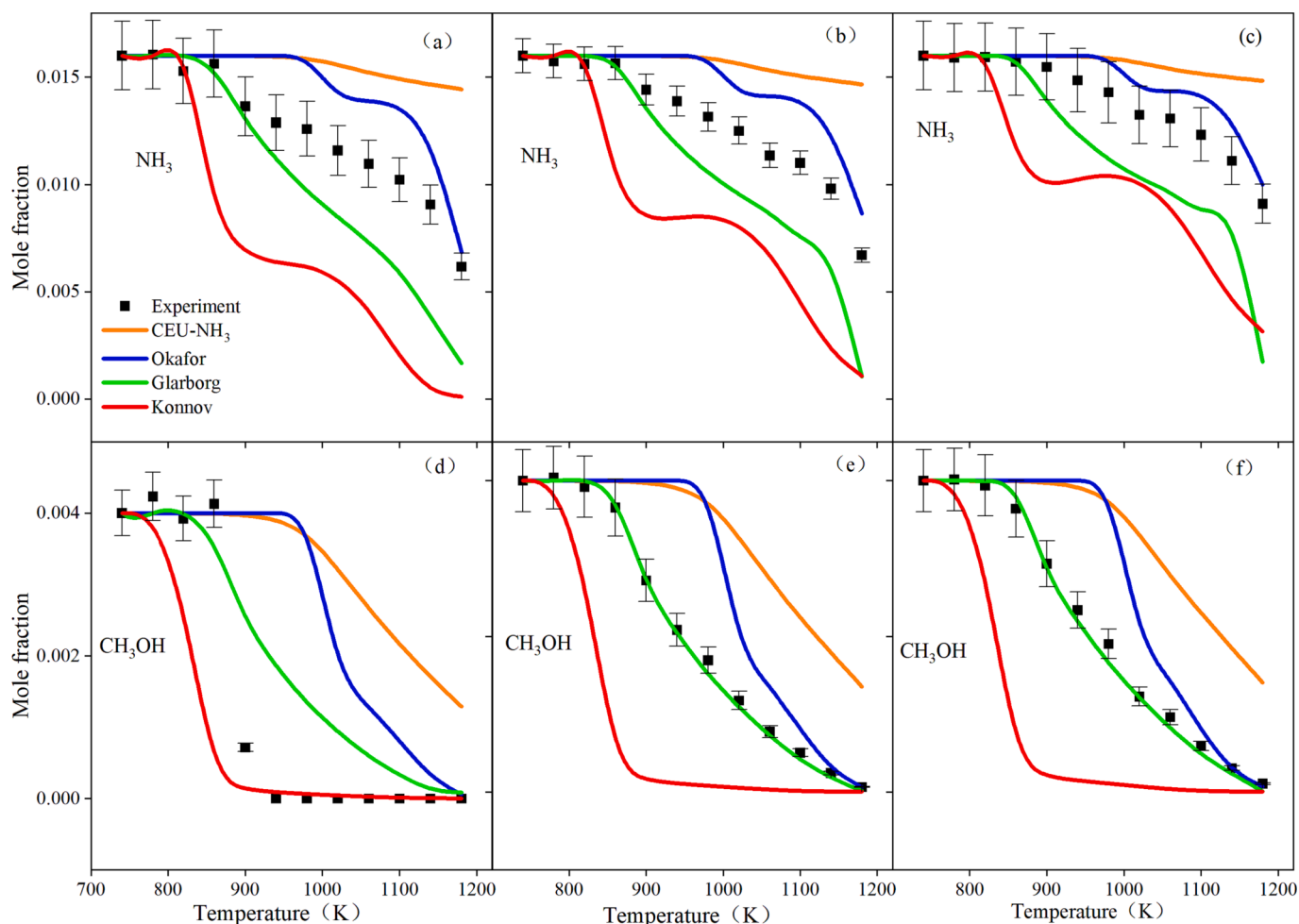


Fig. 2. Comparison of experimental and simulated oxidation of reactants at different equivalence ratios of ammonia/methanol ratio of 4:1. (a) (d) $\phi = 0.8$, (b) (e) $\phi = 1.0$, (c) (f) $\phi = 1.2$. Symbols indicate experimental data and lines indicate simulation results.

Methanol (CH_3OH) is also a potential ammonia combustion enhancer compared to hydrogen, methane, and DME, and can be prepared by reacting green hydrogen (H_2) with CO_2 to form a “carbon-neutral” green fuel. Compared to ammonia, methanol has a rapid laminar flame speed, contains oxygen (which provides oxygen during combustion), is easy to ignite, has a high dilute combustion limit, and is an excellent fuel for spark-ignition internal combustion engines [17]. Using a spherical constant-volume combustion bomb, Liao et al. [18] determined the laminar burning velocity of methanol/air mixtures. At a pressure of 0.1 MPa and a temperature of 358 K, methanol/air flames can attain a maximum burning velocity of 61 cm/s. Vancoillie et al. [19] measured the laminar flame velocity of methanol/air mixtures at 298–358 K and atmospheric pressure using a heat flux method. Experiments revealed that the maximum laminar burning velocity of methanol/air is 57.8 cm/s ($T = 358 \text{ K}$, $\phi = 1.0$). Comparatively, the laminar flame speed of ammonia at room temperature and pressure is only 7 cm/s, indicating that methanol can substantially lift up the laminar burning speed of ammonia/methanol mixtures. In addition, methanol is a carbon-neutral, eco-friendly fuel with mature storage and transportation technologies. Currently, the production of methanol via renewable energy and CO_2 industrialization is becoming increasingly plausible [20–22]. Liu et al. [23] investigated the impact of 2-ethylhexyl nitrate (EHN) on combustion and emissions in a low-load compression ignition engine using pure methanol as the fuel. The results indicated that compared to diesel, the shortened combustion duration of pure methanol was mainly attributed to the reduction in the premixed combustion phase. Wen et al. [24] investigated the

characteristics of methanol active thermal atmosphere combustion (ATAC) using various optical diagnostic techniques on a light-duty engine. Polyoxymethylene dimethyl ether (PODE2) and n-heptane were selected as port fuel injection fuels, while methanol was used as the direct injection fuel. Kinetic analysis demonstrated that even at lower equivalence ratios, PODE2 exhibited lower ignition delay times compared to n-heptane. In addition, ammonia dissolves readily in methanol solution (the ratio of ammonia to methanol is 1:1.13). Schafler et al. [25] used a high-pressure observation method to determine the solubility of ammonia in liquid methanol. They discovered that the molar concentration of ammonia (the mole fraction of ammonia in the liquid) in methanol can reach as high as approximately 66.4 mol·kg⁻¹, meaning that 1 kg of methanol can dissolve 1.13 kg of ammonia. This indicates that at room temperature and pressure, methanol and ammonia can form a stable liquid fuel that is mutually soluble. By combining ammonia and methanol, it is possible to create a zero-carbon, liquid fuel with improved combustion quality that is simple to store, refuel, and transport.

There is limited research on the use of methanol to improve ammonia combustion so far. Wang et al. [26] measured the laminar burning velocity of $\text{NH}_3/\text{CH}_3\text{OH}$ and $\text{NH}_3/\text{C}_2\text{H}_5\text{OH}$ at temperatures ranging from 298 K to 448 K using a heat flux method at atmospheric pressure and compared it to the laminar burning velocity of NH_3/H_2 and NH_3/CH_4 under the same conditions. Lu et al. [27] investigated the chemical reaction mechanism of ammonia/methanol combustion and the effect of adding methanol to ammonia combustion and emissions at various equivalence ratios. Due to the combined effect of high temperature,

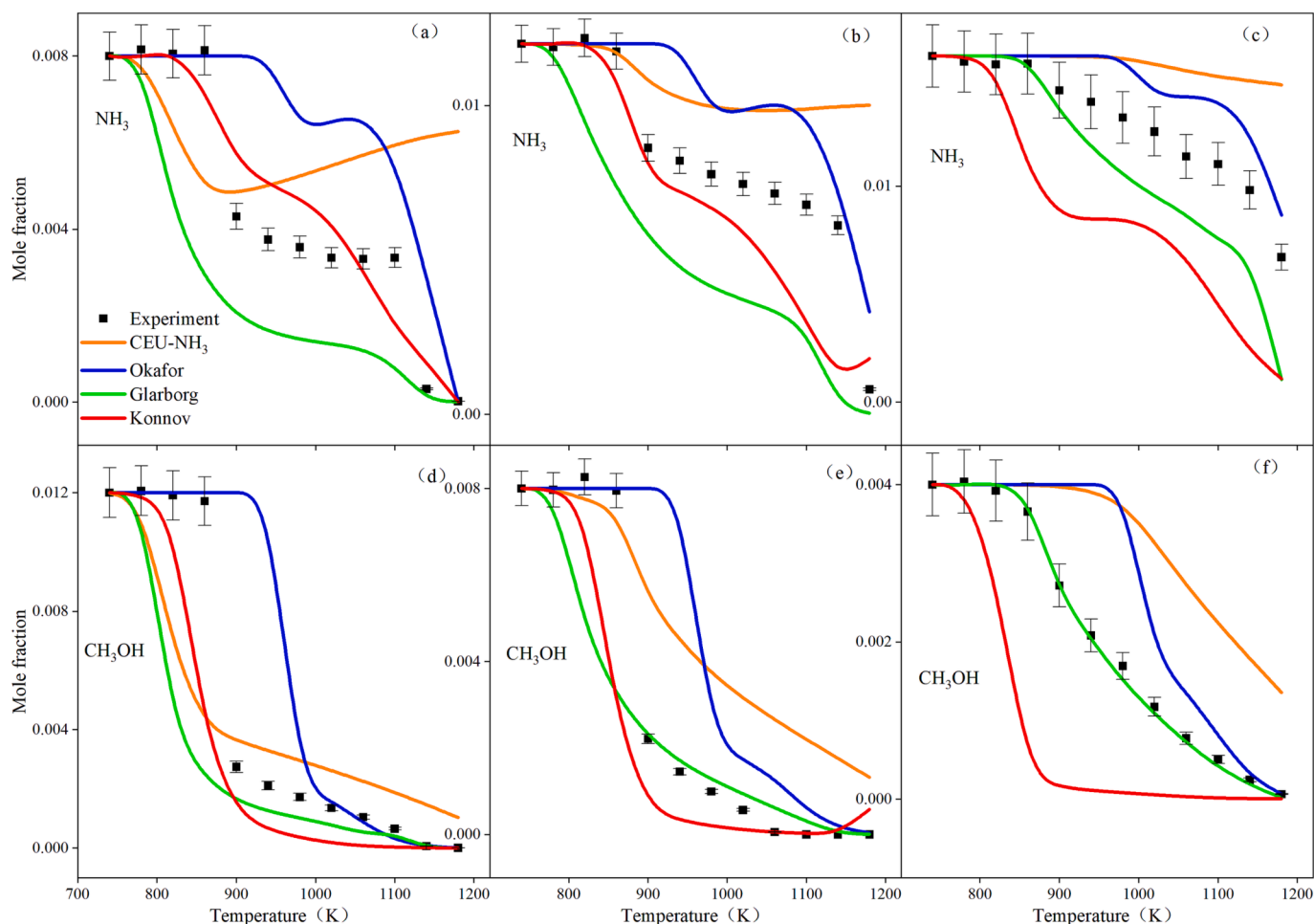


Fig. 3. Experimental and simulated comparison of reactant oxidation at different ammonia to methanol ratios for equivalence ratio $\phi = 1.0$. (a) (d) $\text{NH}_3/\text{CH}_3\text{OH} = 2:3$, (b) (e) $\text{NH}_3/\text{CH}_3\text{OH} = 3:2$, (c) (f) $\text{NH}_3/\text{CH}_3\text{OH} = 4:1$. Symbols indicate experimental data and lines indicate simulation results.

excess oxygen, and accumulated HO_2 free radicals, the addition of methanol increases NO production, as shown in [26]. In a high-pressure shock tube (HPST), Nadiri et al. [28] measured the ignition delay time (IDT) of ammonia/methanol mixtures and found that adding a very low concentration of methanol to ammonia significantly shortened the IDT. In addition, they generated the kinetic mechanism of ammonia/methanol combustion with the software reaction mechanism generator (RMG) and validated the mechanism with previously collected experimental data. Li et al. [29] also investigated the IDT of ammonia/methanol mixtures utilizing HPST and improved the kinetic mechanism of ammonia/methanol co-oxidation. They found that adding a small amount of methanol can significantly reduce the IDT of the ammonia/methanol mixture, and adding 5% methanol can reduce the IDT by over 60%.

Current research on the combustion of ammonia/methanol mixtures is limited. As a foundation for investigating the co-combustion of ammonia/methanol, there is a scarcity of experimental data on the parent compounds, intermediates and final products during the process of ammonia/methanol co-oxidation. The reaction kinetics mechanism is also inaccurate and requires refinement based on experimental foundations. Given this context, the purpose of this study is to use the SVUV photoionization mass spectrometry technique to conduct oxidation experiments of $\text{NH}_3/\text{CH}_3\text{OH}$ in a JSR and to measure the key intermediates and products during the oxidation process. Several published kinetic mechanism models for packages of NH_3 and CH_3OH are compared with experimental results and optimized on this basis, and the intermediate processes of ammonia/methanol fuel blends are also analyzed.

2. Experimental methods and chemical kinetic modelling

2.1. Experimental setup

The ammonia/methanol oxidation experiment was conducted at the BL09U Atomic and Molecular Physics Station at the University of Science and Technology of China's National Synchrotron Radiation Laboratory (NSRL). As depicted in Fig. 1, the experimental apparatus consists of a time-of-flight molecular mass spectrometer and a chromatograph utilizing synchrotron radiation photoionization technology in conjunction with a JSR reactor. The experimental apparatus has five components: the injection system, the gas supply system, the heating system, the reaction system, and the analysis system. The experimental procedure is as follows: liquid methanol is delivered to the vaporization tank via a liquid flow controller, and the vaporization tank is maintained at a temperature just above the methanol's boiling point. After evaporation, methanol is carried by ultra-pure argon gas, and ammonia gas is introduced directly from the ammonia gas cylinder. The two gases are combined and then fed into the tube furnace-heated reactor. By controlling the reaction temperature, flow rate, pressure, and other variables, the desired experimental conditions are attained.

The injection system consists of an injection needle (Hamilton 50 mL) and a precision syringe pump (LSP01-2A, Baoding Langhe Constant Flow Pump Co., Ltd.). With an accuracy of 0.01 L/min, the injection pump controls the sample delivery rate via the injection needle. A U.S.-made MKS flow meter regulates the gas mass flow. In the experiment, one path of argon gas (fixed flow rate of 200 sccm) was used as the

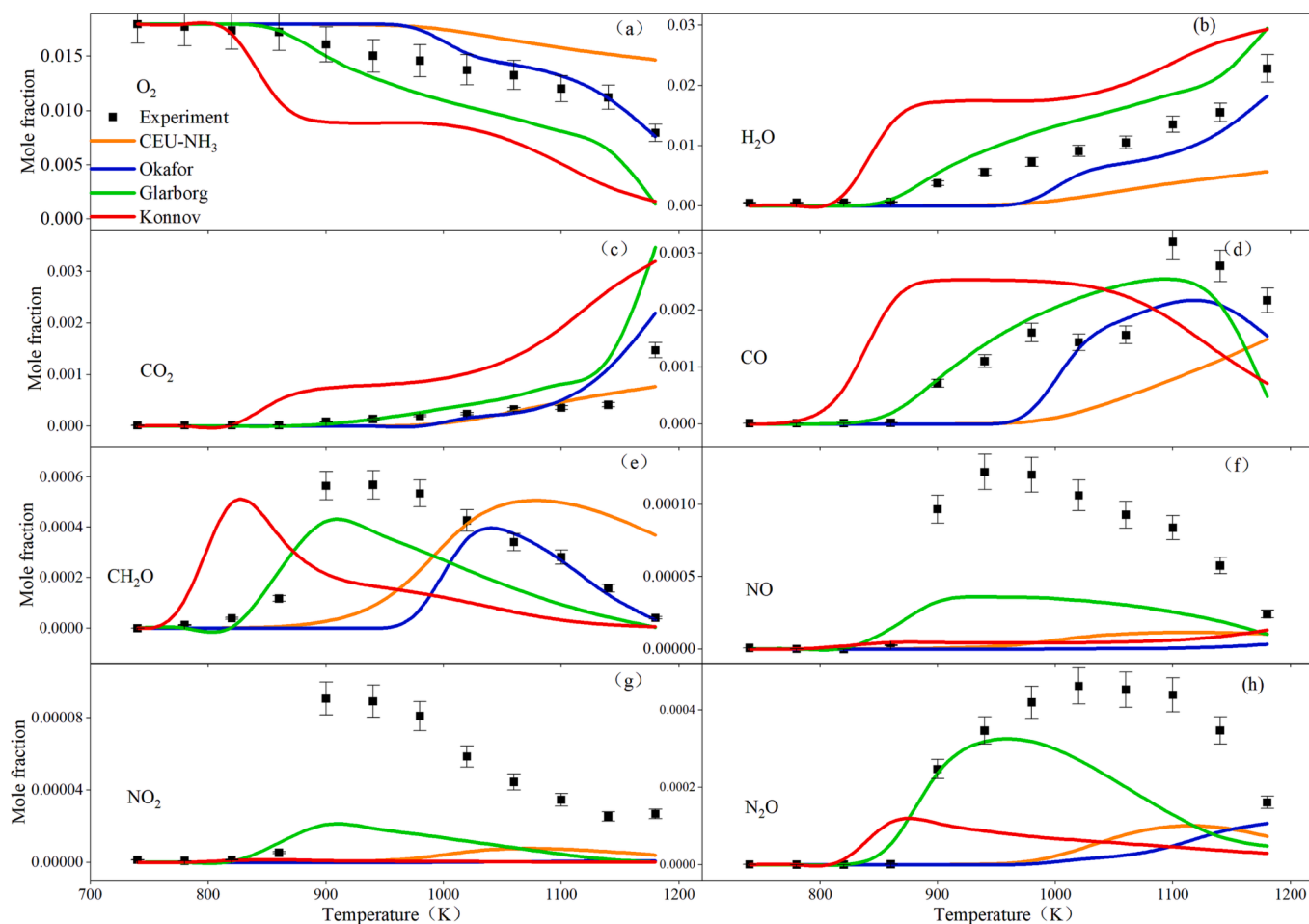


Fig. 4. Experimental and simulation comparison of intermediates and final products at an ammonia/methanol ratio of 4:1 at an equivalence ratio of 1.0.

Table 2

Reaction rate constants for some important reactions in the ammonia/methanol mechanism, in (s^{-1} , cm^3 , J/mol).

NO	Selected reactions	A	n	Ea	Ref.
R1	$\text{NH}_3 + \text{H} = \text{NH}_2 + \text{H}_2$	6.4E05	2.390	10,171	[34]
R2	$\text{NH}_3 + \text{O} = \text{NH}_2 + \text{OH}$	9.4E06	1.940	6460	[34]
R3	$\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$	2.0E06	2.040	566	[34]
R4	$\text{CH}_3\text{OH} + \text{OH} = \text{CH}_2\text{OH} + \text{H}_2\text{O}$	1.5E08	1.4434	113	[34]
R5	$\text{HCO} + \text{O}_2 = \text{CO} + \text{HO}_2$	6.9E06	1.900	-1369	[34]
R6	$\text{HNO} + \text{O}_2 = \text{HO}_2 + \text{NO}$	2.0E13	0	16,000	[34]
R7	$\text{H}_2\text{NO} + \text{O}_2 = \text{HNO} + \text{HO}_2$	2.3E02	2.994	16,500	[34]
R8	$\text{CH}_3\text{O}(+\text{M}) = \text{CH}_2\text{O} + \text{H}(+\text{M})$	1.1E15	1.210	24,069	This work
R9	$\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$	8.0E18	-2.369	870	This work
R10	$\text{NH}_2 + \text{HONO} = \text{NH}_3 + \text{NO}_2$	7.1E11	3.020	-4940	This work
R11	$\text{H}_2\text{NO} + \text{NO}_2 = \text{HONO} + \text{HNO}$	4.4E04	2.640	4040	This work
R12	$\text{NO}_2 + \text{H} = \text{NO} + \text{OH}$	1.8E12	0	362	This work
R13	$\text{H}_2\text{NN} + \text{O}_2 = \text{NH}_2 + \text{NO}_2$	1.5E06	0	5961	This work
R14	$\text{CH}_3\text{OH} + \text{NO}_2 = \text{HONO} + \text{CH}_2\text{OH}$	1.5E12	3.320	20,035	This work

carrier gas and entered the gasification tank with the experimental reagents. The vaporized reagent gas was combined with a second path of argon gas (diluent gas) and introduced into the target reactor (JSR). The heating system comprises a thermocouple of type K, a temperature

controller, and several heating belts. The heating belts are wrapped tightly around the surface of each pipeline, then covered with aluminum foil and cotton thermal insulation. This prevents direct contact with air, reduces heat loss, and ensures uniform pipeline heating. On each pipeline and on the interior and exterior of the gasification tank, thermocouples are installed to measure the actual temperature of the pipeline surface in real time. The reaction system's spherical reactor is composed of silica, which can prevent wall reactions. The uncertainty of the reaction temperature was within ± 10 K (The experimental error comes from the nominal error of K-type thermocouples with models KW/KB/KS.). The reactor has a volume of 78 cm^3 and is composed of an inner and an outer tube. The outer tube has a diameter of 25.4 mm, while the inner tube has an outer diameter of 6 mm and an inner diameter of 2 mm with a volume of 0.245 cm^3 . Four opposing nozzles serve as the reactor's inlet and, through jet stirring, can achieve perfect and uniform mixing. In the experiment, the fuel's residence time was set to 2 s, and the temperature range was 750–1,200 K. The inlet pipe has an inner diameter of 0.3 cm, while the sampling probe has an inner diameter of 0.1 cm. The final product of the experiment was analyzed by mass spectrometry after being sampled through a quartz cone-shaped nozzle. Through the tip of a quartz microprobe, another route was extracted and pumped into gas chromatography for analysis. In this paper, the introduction to the synchrotron vacuum ultraviolet ionization beamline is based on the work of Wang, Zhou, and others [30,31].

In this pyrolysis experiment, the reaction residence time was 2 s, the pressure was 760 Torr, and the carrier and diluent gases were both argons. The mole quantities of each inlet fuel are displayed in Table 1, and they were managed by an MKS mass flow controller. In these five operating conditions, the following observations can be made: a

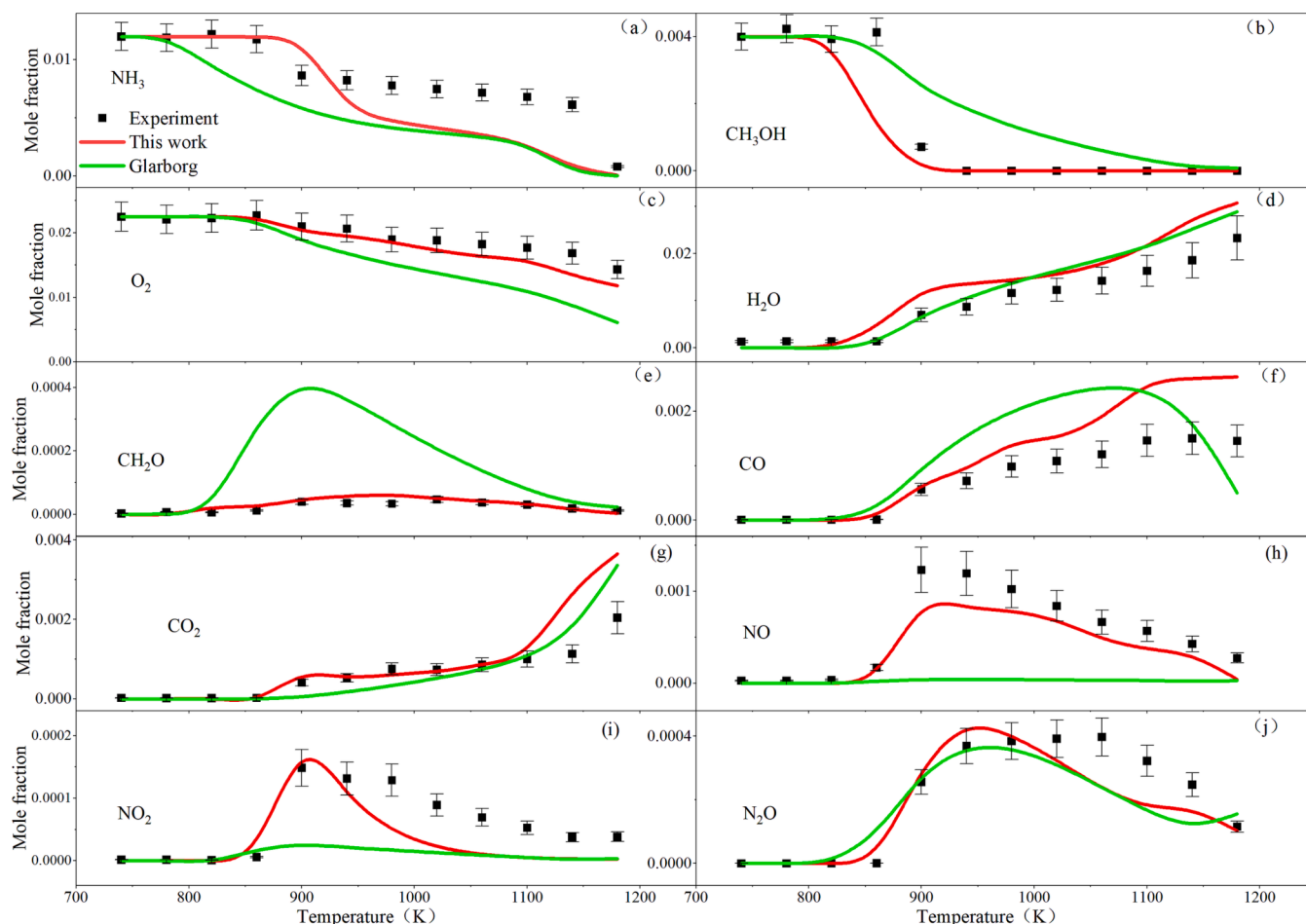


Fig. 5. Comparison of reactants, intermediates and final products between experiments and simulations with optimized mechanism at an ammonia/methanol ratio of 4:1 at an equivalence ratio $\phi = 1.0$.

comparison of experimental results with different ammonia/methanol ratios at the same equivalence ratio, and a comparison of experimental results with different equivalence ratios at the same ammonia/methanol ratio. In order to maintain a constant residence time, the gas flow rate was adjusted based on the temperature of the reactor. The experiment identified and quantified species with ionization energies in the range of 11.5–14.5 eV and a predetermined temperature range of 740 K–1180 K. The uncertainty of the mole fraction calibrated is within $\pm 10\%$ for both SVUV photoionization mass spectrometry and GC analysis. The experimental error is provided by the laboratory. The overall error includes contributions from equipment such as time-of-flight molecular mass spectrometer, chromatograph, and JSR reactor. For specific information, please refer to the referenced literature [32,33].

2.2. Materials

CH₃OH was provided by China National Pharmaceutical Group Chemical Reagent Co., Ltd., with a purity (GC) greater than 99.5%. The purities of NH₃, O₂, and Ar were all greater than 99.999%. High-purity argon gas (99.9995%) was used as the diluent and carrier gas. The oxygen flow rate was controlled at 1% using an MKS mass flow controller.

3. Chemical kinetic model for the oxidation of ammonia/methanol

JSR experiments were simulated using the ANSYS Chemkin 17.0 PSR (perfectly stirred reactor) module. The fundamental simulation parameters used in PSR simulation were consistent with those of the JSR

experiments. In this study, four commonly used ammonia/methanol mechanisms were compared, including CEU-NH₃ [26], Glarborg [34], Konnov [35], and Okafor [9]. Wang et al. [26] proposed the CEU-NH₃ mechanism for ammonia/methanol combustion based on the published experimental data. The results of this mechanism were consistent with the published data on ignition delay time and laminar burning velocity. Glarborg et al. [34] developed a comprehensive mechanism for the ammonia reaction based on previous nitrogen chemistry research, which has been the most prevalent mechanism for ammonia. Konnov et al. [35] proposed a revised detailed reaction mechanism for the combustion of small hydrocarbons that includes reaction kinetics models for ammonia/methanol mixtures and makes accurate predictions for ammonia reaction processes. Okafor et al. [9] developed a comprehensive ammonia reaction mechanism and experimentally validated it. As this mechanism has good predictions for the combustion of ammonia and other fuel mixtures, it was also selected for the comparison of simulation data and experimental results. Figs. 2 and 3 show the consumption trends of NH₃ and CH₃OH during the NH₃/CH₃OH oxidation experiments, along with a comparison to the simulation results of various chemical kinetics models. By comparing the results, it was determined that the simulation results can be roughly categorized into three groups, with the Konnov model having the highest reaction activity, the Okafor and Glarborg models falling in the middle, and the CEU-NH₃ model having the lowest reaction activity. Specifically, in experiments with a high ammonia concentration, as depicted in Fig. 2, ammonia and methanol began to oxidize significantly at approximately 860 K, which was 40 K earlier than the oxidation experiment of pure ammonia in the previous study [36]. However, the Konnov model did

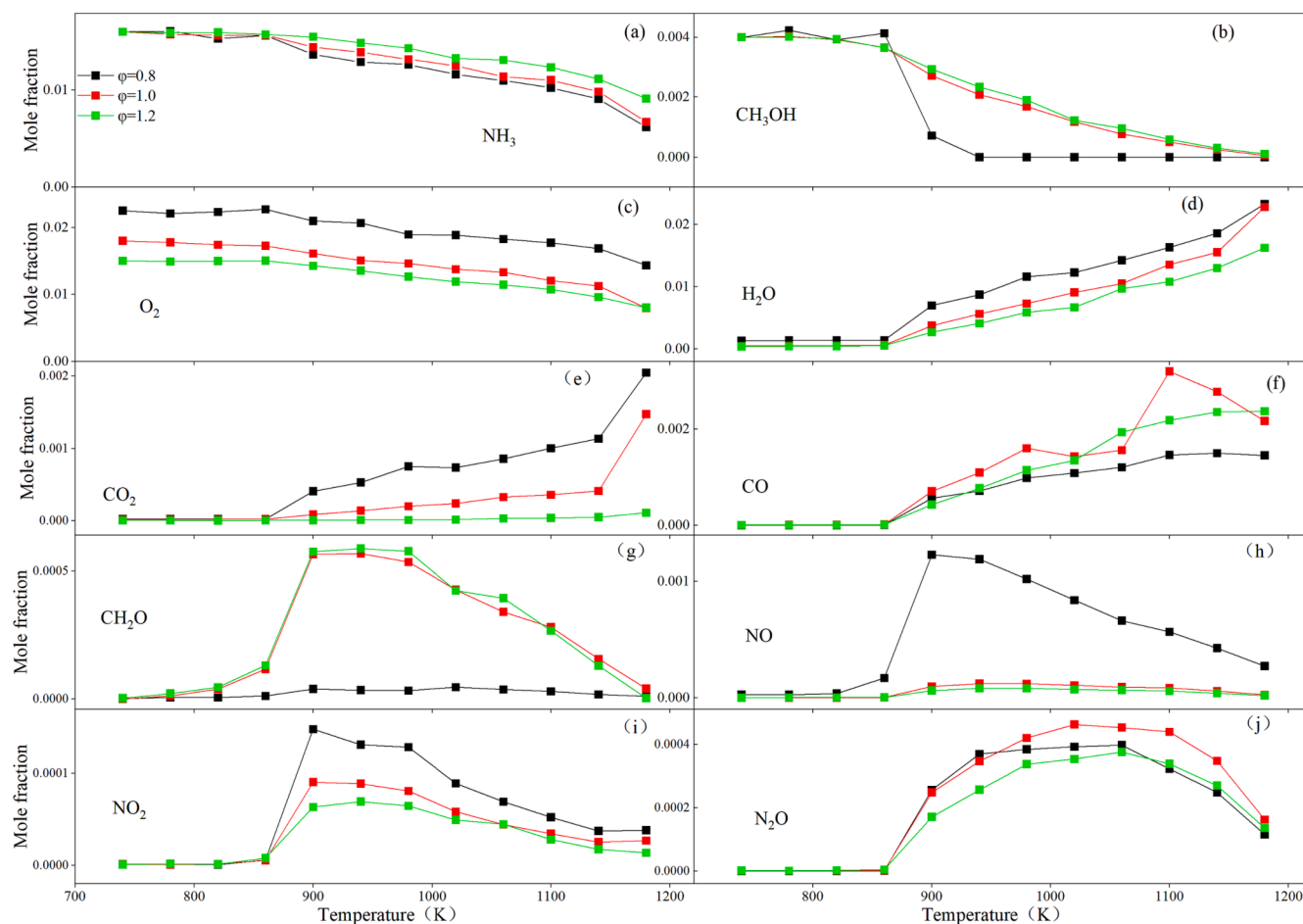


Fig. 6. Experimental results at different equivalence ratios of ammonia/methanol at 4:1.

not accurately predict the temperature, and the CEU-NH₃ and Okafor models overestimated it. Existing models did not accurately predict the oxidation rate of ammonia and methanol, with the CEU-NH₃ model predicting a lower oxidation rate and the Konnov model predicting a higher rate. As depicted in Fig. 3, under different ammonia/methanol conditions with equivalence ratio of 1.0, the Glarborg [34] model had a good prediction for CH₃OH consumption but a less-than-ideal prediction for NH₃ consumption at a relatively low ammonia/methanol ratio of 2:3. When the ratio of ammonia to methanol was relatively high at 4:1, only the Glarborg model accurately predicted the consumption of CH₃OH and NH₃. In general, the majority of chemical kinetic models appear to overestimate the NH₃ oxidation rate, i.e., they predict a rapid decrease in the oxidation rate after the beginning of oxidation. Through experimental verification, it has been determined that the rate of ammonia oxidation slows down as the oxidation process progresses.

Fig. 4 compares the experimental and simulated molar fractions of intermediates and end products under conditions of a 1.0 equivalence ratio and ammonia/methanol ratio of 4:1. It can be observed that all models predict O₂ and the final products (H₂O and CO₂) with reasonable accuracy, but there are substantial discrepancies in the prediction of intermediates. The Konnov model overestimates CO and underestimates the temperature at which CH₂O begins to form. The CEU-NH₃ model predicts lower molar fraction results for CO but overpredicts the temperature at which CH₂O starts to form; the Okafor model predicts CO more accurately but overpredicts the temperature at which CH₂O starts to form. The Glarborg model is more consistent for CO and CH₂O. The prediction of nitrogen oxides (NO, NO₂, N₂O) is, however, underestimated by all mechanisms. Clearly, the current combustion mechanisms do not adequately predict the process of ammonia/methanol co-

combustion, necessitating additional research and optimization. Moreover, given the relatively good predictive performance of the Glarborg model for the entire oxidation process, this study opts to further optimize the Glarborg model [34] in order to enhance the prediction of experimental results.

The complete NH₃/CH₃OH mechanism is based on the mechanism established by Glarborg et al. [34] and includes 5 elements (C, H, O, N), 1397 reactions, and 151 species. The kinetic model consists of four subsets: (1) C₁-C₂ hydrocarbon subset (detailed chemical kinetic models for C₁-C₂ under high-temperature and high-pressure conditions were established by Hamid Hashemi et al. [37,38] and Jorge Jimenez-Lopez et al. [39] and experimentally validated), (2) amine subset (Stephen J. Klippenstein et al. [40] extensively studied the influence of amines on thermal denitrification and developed a chemical kinetic model for it), (3) cyanide subset (Philippe Dagaut et al. [41] summarized the formation pathways of HCN and the thermochemical properties of related species, and established the chemical kinetic reaction mechanism for HCN oxidation), and (4) hydrocarbon/nitrogen interaction subset (Peter Glarborg et al. [42] and Teresa Mendiara [43,44] established a chemical kinetic model for hydrocarbon-nitrogen reactions, which was validated through experiments). The physical parameters of the major components involved in the oxidation of NH₃/CH₃OH mixture were obtained through quantum chemical calculations, and all reactants and transition states were calculated using Gaussian 16 [45] density functional theory. The reaction rate parameters for the relevant reaction pathways are shown in Table 2.

From the perspective of the reaction process, NH₃ oxidation begins with the dehydrogenation of NH₂, which is predominantly produced by the reactions NH₃ + H = NH₂ + H₂ (R1), NH₃ + O = NH₂ + OH (R2), and

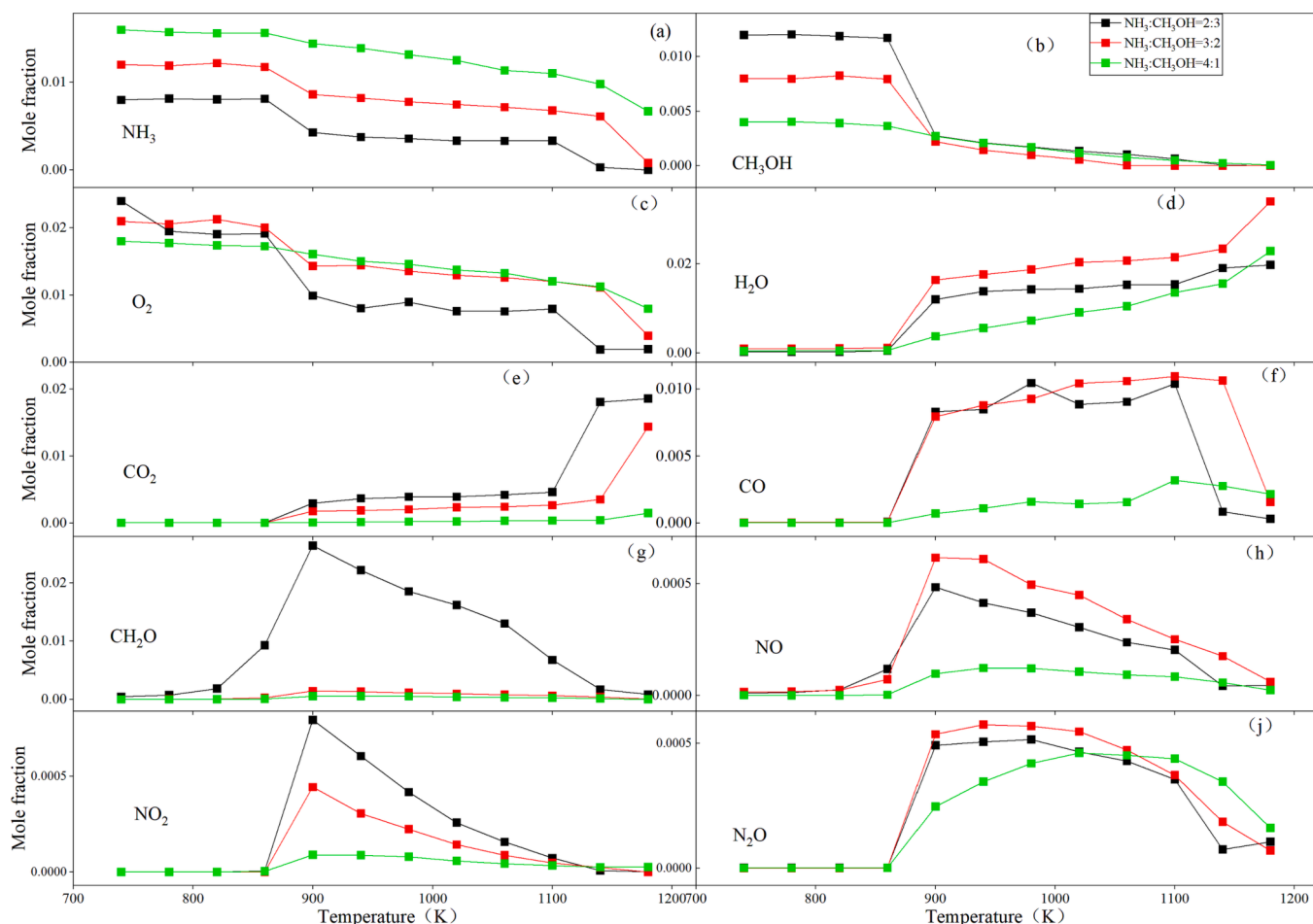


Fig. 7. Experimental results for different ammonia/methanol ratios at the equivalent ratio $\phi = 1$.

$\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$ (R3). NH_2 reacts with nitrogen oxides and hydrocarbons to produce several essential intermediates. Most of the CH_3OH is consumed by $\text{CH}_3\text{OH} + \text{OH} = \text{CH}_2\text{OH} + \text{H}_2\text{O}$ (R4), and the resulting CH_2OH reacts with H , OH , and O_2 to produce CH_2O . Then, CH_2OH and CH_2O react with other substances to generate HCO , and $\text{HCO} + \text{O}_2 = \text{CO} + \text{HO}_2$ (R5) generates the majority of CO . HO_2 is a highly oxidizing compound that plays a vital role in the reaction. $\text{HNO} + \text{O}_2 = \text{HO}_2 + \text{NO}$ (R6) and $\text{H}_2\text{NO} + \text{O}_2 = \text{HNO} + \text{HO}_2$ (R7) are two significant reactions that generate HO_2 . HO_2 produces a substantial amount of OH , which facilitates the dehydrogenation of NH_3 and CH_3OH and drives the reaction. The reaction rate coefficients for R1-R7 are derived from the mechanism of Glarborg et al. [34], while the focus is to modify the reactions involving nitrogen oxides (R8-R14) due to the inaccurate prediction of the original mechanism.

Fig. 5 depicts the molar fraction distributions of reactants, intermediates, and final products in experiments with a 4:1 ammonia/methanol ratio and a 1:0 equivalence ratio, as well as the simulated results of the optimized mechanism (named NGH-W) and Glarborg et al.'s [34] mechanism. Compared to Glarborg's mechanism, the optimized mechanism has made significant improvements in predicting intermediates and nitrogen oxides, and the results are in close agreement with experimental data.

4. Results and discussion

4.1. Experimental results

Figs. 6 and 7 depict the distribution of reactants, intermediates, and products during the ammonia/methanol oxidation process at various

equivalence ratios and ammonia/methanol ratios. In all five operating conditions, ammonia begins to oxidize at a temperature of approximately 860 K, and the oxidation rate increases with temperature until it abruptly slows at approximately 900 K. In contrast, under operating conditions with an equivalence ratio of 1.0, an ammonia/methanol ratio of 2:3, and a temperature of 1100 K, the oxidation rate increases until all NH_3 is oxidized. In addition, as shown in Fig. 7, as the ratio of ammonia to methanol increases, so does the temperature at which the rate of oxidation is accelerating again occurs. Similar to ammonia, methanol begins to oxidize at approximately 860 K, but its oxidation rate is significantly faster. It has been observed that the majority of methanol oxidation occurs before 1000 K. At a ratio of 4:1 ammonia to methanol ratio and an equivalence ratio of 1.0 and 1.2, the oxidation characteristics of methanol become remarkably similar, with both decreasing monotonically. At an equivalence ratio of 0.8, however, the consumption rate of methanol is significantly higher than that at equivalence ratios of 1.0 and 1.2, indicating that methanol's oxidation rate is rapid and temperature-sensitive under high oxygen concentrations. In addition, methanol is completely oxidized at approximately 940 K, whereas ammonia continues to oxidize at 1100 K.

CO and CH_2O are important intermediates during the thermal decomposition of CH_3OH . The equivalence ratio has a significant effect on the production of CO . As shown in Fig. 6, when the ammonia/methanol ratio is 4:1, the trend of increasing CO molar fraction under different equivalence ratios is similar. At an equivalence ratio of 1, the molar fraction of CO is at its greatest. Specifically, as shown in Fig. 7, at an equivalence ratio of 1.0 and an ammonia/methanol ratio of 2:3 and 3:2, the trend of molar fraction of CO is roughly the same, that CO formation starts at approximately 860 K, slows at approximately 900 K,

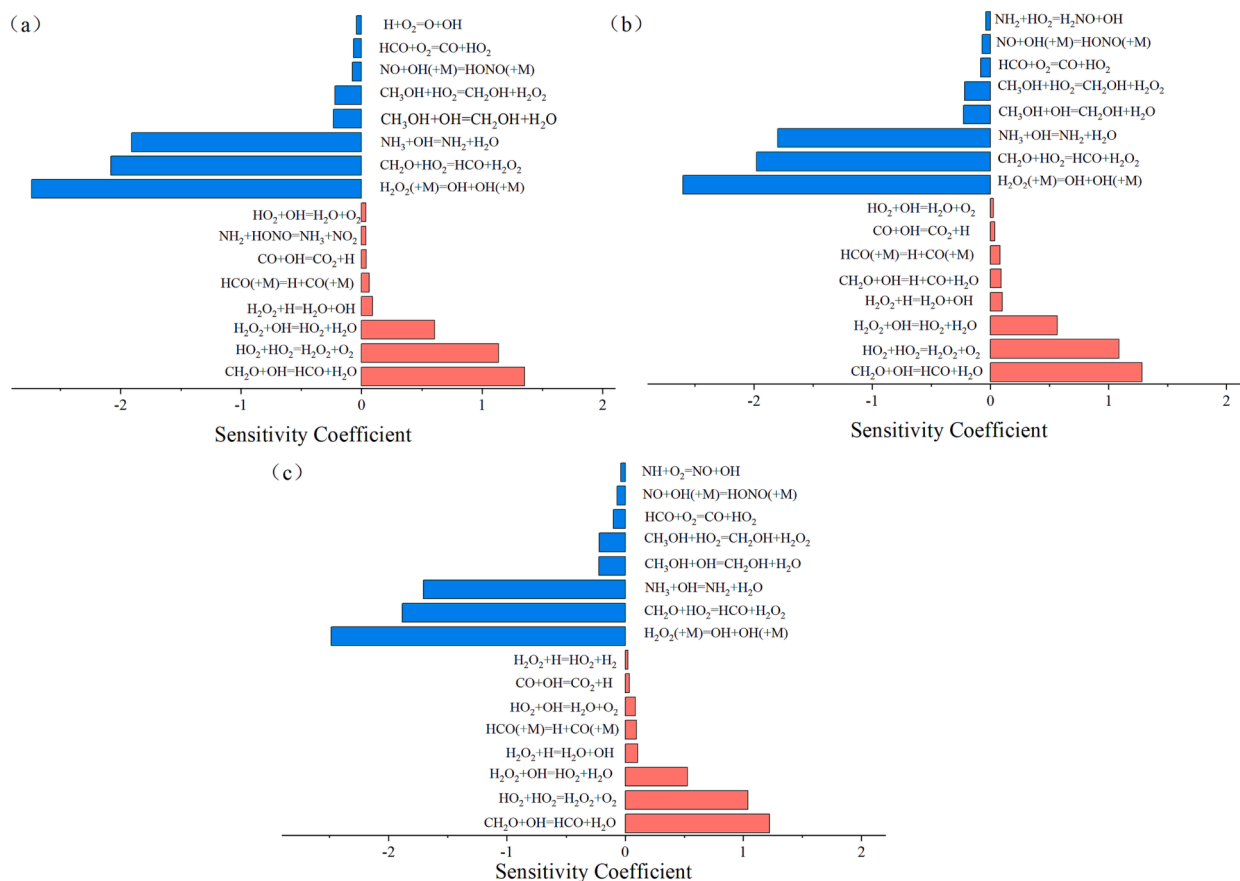


Fig. 8. Sensitivity analysis of methanol at 860 K for different equivalence ratios at ammonia/methanol ratio of 4:1. (a) $\phi = 0.8$, (b) $\phi = 1.0$, (c) $\phi = 1.2$. Positive values indicate inhibition effect and negative values indicate promotion effect.

and abruptly decreasing after 1000 K. When the methanol concentration is high, the majority of CO is produced at temperatures between 900 K and 1100 K.

Formaldehyde (CH_2O) is another significant intermediate product of combustion. The analysis of the reaction pathways of ammonia/methanol by Li et al. [28] reveals that CH_3OH oxidation generates CH_2OH and CH_3O , and CH_2OH and CH_3O further form CH_2O , which forms HCO as the reaction terminates. At a 4:1 ammonia/methanol ratio and an equivalence ratio between 1.0 and 1.2, the molar fraction of CH_2O is significantly higher than that at an equivalence ratio of 0.8, and reaches a maximum at approximately 900 K. It is evident that less CH_2O is produced under oxygen-rich conditions. As depicted in Fig. 3, at an equivalence ratio of 1.0, the molar fraction of CH_2O under a 2:3 ammonia/methanol ratio is significantly greater than under ratios 3:2 and 4:1. CH_2O production begins at 800 K, peaks at approximately 950 K, and is completely consumed at approximately 1100 K.

Nitrogen oxides (NO, NO_2 , and N_2O) are generally produced at higher temperature. At a ratio of 4:1 ammonia/methanol, the mole fraction of NO reaches its maximum around 920 K and then decreases gradually with increasing temperature, as shown in Fig. 6. The mole fraction of NO is significantly greater at an equivalence ratio of 0.8 than at equivalence ratios of 1.0 and 1.2. This indicates that under oxygen-rich conditions, more NO is produced. Under conditions of low oxygen, the formation of HO_2 is inhibited, and the reactions $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ and $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$ are inhibited, resulting in an increase in NO with an increase in oxygen [9]. As depicted in Fig. 7, the highest NO production is observed at an ammonia/methanol ratio of 3:2 and an equivalence ratio of 1.0, followed by an ammonia/methanol ratio of 2:3, and the lowest is observed at an ammonia/methanol ratio of 4:1, with a total production of only 1.2×10^{-4} . Similar to NO, the mole

fraction of NO_2 at a 4:1 ammonia/methanol ratio and an equivalence ratio of 0.8 is greater than that at equivalence ratios of 1.0 and 1.2. However, the mole fractions are similar across equivalence ratios, indicating that oxygen has limited effect on its production. The highest NO_2 production is observed at ammonia to methanol ratio of 2:3 and an equivalence ratio of 1.0, followed by ammonia to methanol a ratio of 3:2 and 4:1. At an ammonia/methanol ratio of 4:1, the mole fraction of N_2O exhibits a similar trend at various equivalence ratios, beginning to form at approximately 860 K, reaching its maximum at approximately 1000 K, and then gradually decreasing. It can be seen in Fig. 7 that the molar fraction trend of N_2O for different ammonia/methanol ratios at the same equivalent ratio is similar to the molar fraction trend of N_2O , as shown in Fig. 6. This suggests that different equivalence ratios and ammonia/methanol ratios have little effect on the mole fraction of N_2O .

H_2O and CO_2 , the products corresponding to the reactants, begin to form at approximately 860 K and increase gradually as the temperature rises. The molar fraction of H_2O is the greatest when the ammonia/methanol ratio is 4:1 and the equivalence ratio is 0.8, followed by equivalence ratios of 1.0 and 1.2. When the equivalence ratio is fixed at 1.0, as shown in Fig. 7, the molar fraction of H_2O is greatest at an ammonia/methanol ratio of 3:2, followed by 2:3 and 4:1. Maximum CO_2 production occurs at an equivalence ratio of 0.8 when the ammonia/methanol ratio is 4:1, followed by equivalence ratios of 1.0 and 1.2. At an equivalence ratio of 1.0, the molar fraction of CO_2 is highest at ammonia to methanol ratio of 2:3 and lowest at a ratio of 4:1. At low equivalence ratios, a high concentration of oxygen favors the formation of HO_2 , which generates a large number of OH radicals, thereby promoting the oxidation of ammonia and methanol. At low equivalence ratios, the molar fraction of H_2O and CO_2 is consequently greater.

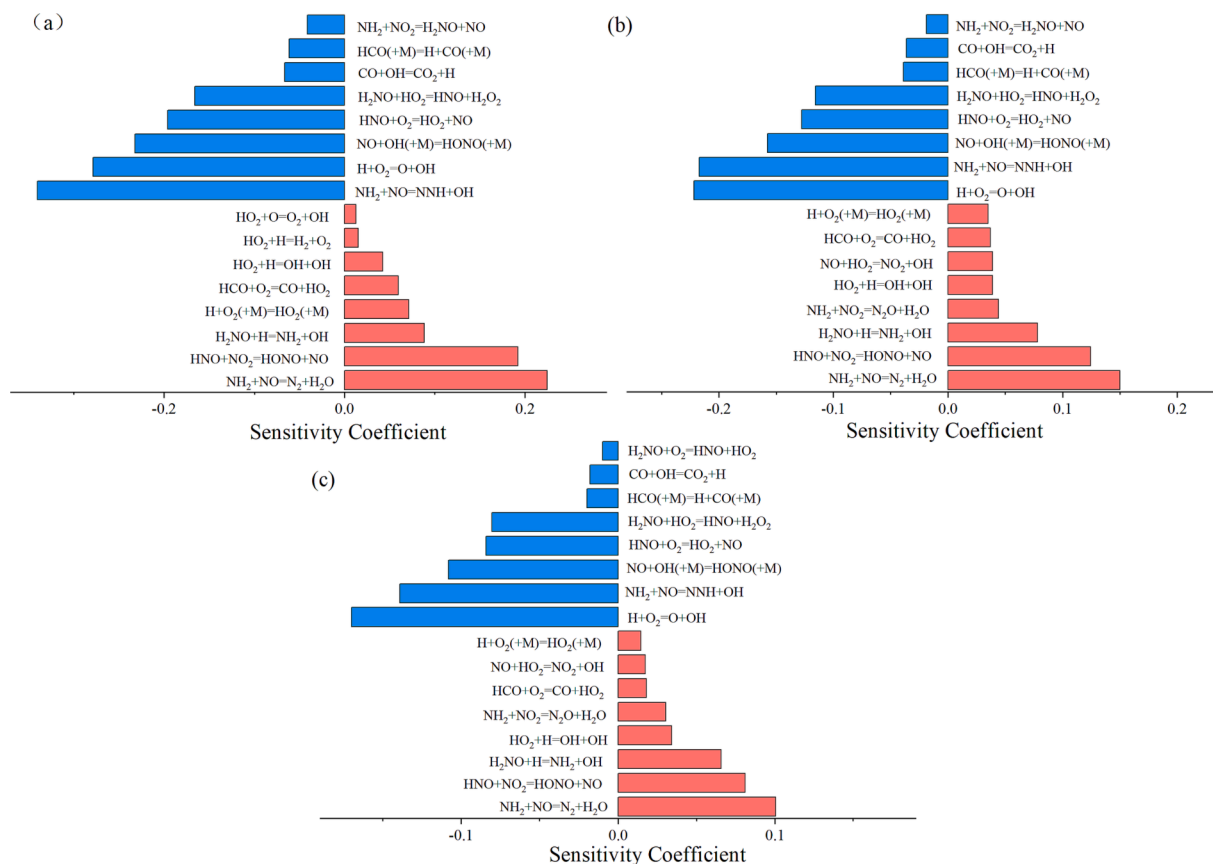


Fig. 9. Sensitivity analysis of ammonia at 1100 K for different equivalence ratios at an ammonia/methanol ratio of 4:1. (a) $\phi = 0.8$, (b) $\phi = 1.0$, (c) $\phi = 1.2$. Positive values indicate inhibition effect and negative values indicate promotion effect.

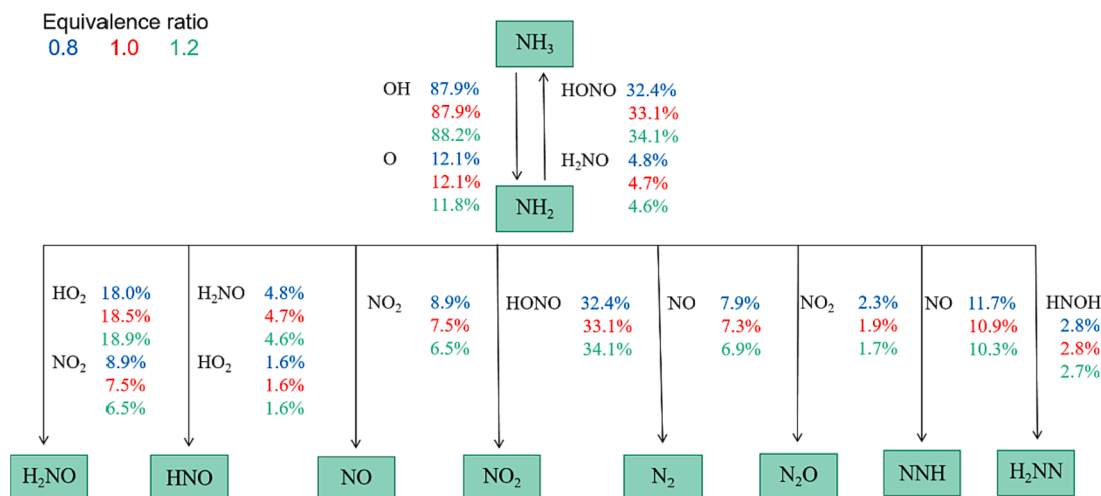


Fig. 10. Pathway of ammonia oxidation at 1100 K with ammonia/methanol ratio of 4:1 at $\phi = 0.8/1.0/1.2$.

4.2. Sensitivity analysis under different equivalence ratios

In order to analyze the reaction process of ammonia and methanol oxidation in greater detail, a sensitivity analysis was performed on the parent at 760 torr using the improved reaction mechanism. Since ammonia and methanol react significantly at different temperatures, ammonia sensitivity analysis was performed at 1100 K and methanol sensitivity analysis was performed at 860 K. This is due to the fact that at these two conditions, the molar fractions of the majority of intermediate products reach their maximum values, which is more conducive to a

thorough analysis of the oxidation process. Negative sensitivity coefficients indicate that the reaction is promoted, whereas positive sensitivity coefficients indicate that the reaction is inhibited. Figs. 8 and 9 depict the sixteen intermediate reactions with the greatest impact on methanol and ammonia at various equivalence ratios. It can be seen that the reaction $\text{H}_2\text{O}_2(+M) = \text{OH} + \text{OH}(+M)$ has the greatest impact on the methanol consumption rate. This reaction will generate a substantial amount of the reactive free radical OH, which will facilitate the consumption of CH_3OH in high-temperature oxidation. The second most significant reaction is $\text{CH}_2\text{O} + \text{HO}_2 = \text{HCO} + \text{H}_2\text{O}_2$, in which CH_2O is

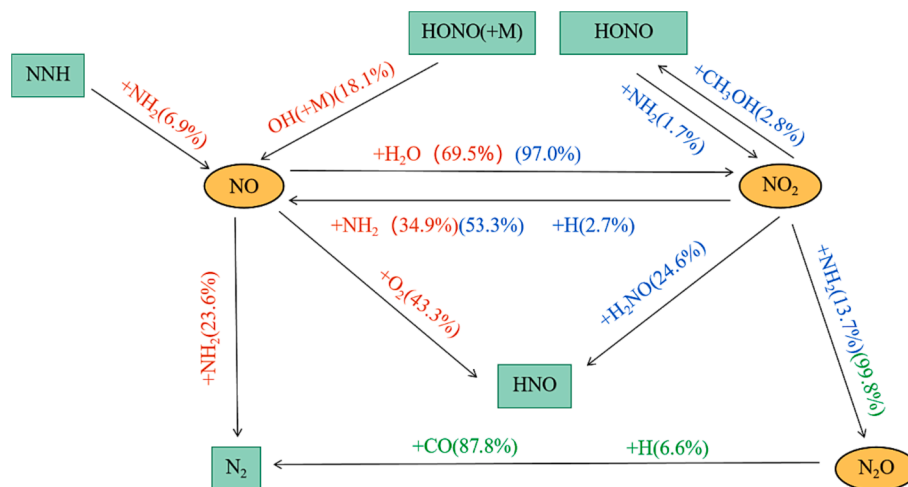


Fig. 11. NO_x conversion pathway at 1100 K for an ammonia/methanol ratio of 4:1 and $\phi = 1.0$, with red values for NO, blue values for NO₂, and green values for N₂O. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

oxidized to HCO while H₂O₂ is produced. The majority of CH₃OH is converted to CH₂O, a crucial intermediate, and the CH₂O reaction also generates HCO, which ultimately reacts to form CO₂. Notably, NH₃ + OH = NH₂ + H₂O has a similar effect on CH₃OH as the previous two reactions, indicating that NH₃ can promote the oxidation of CH₃OH by reacting with the free radical OH to generate NH₂ at relatively high ammonia/methanol ratios. Three reactions exert the greatest influence on NH₃: NH₂ + NO = NNH + OH, H + O₂ = O + OH, and NO + OH(+M) = HONO(+M). These reactions are all found to be related to the free radical OH. At the same time, it is observed that NH₂ + NO has two major branches, namely NH₂ + NO = NNH + OH and NH₂ + NO = N₂ + H₂O, one of which has a stimulatory effect on NH₃, while the other has an inhibitory effect on NH₃. However, both reactions are weakened at high equivalence ratios.

4.3. Generation of nitrogen oxides

The pathway analysis of ammonia oxidation at 1100 K with a 4:1 ammonia/methanol ratio and 0.8/1.0/1.2 equivalence ratio is depicted in Fig. 10. It can be observed that NH₂ produced from NH₃ reacts predominantly with NO_x and HO₂. The NH₂ + HONO = NH₃ + NO₂ reaction, which is the primary route for NH₃ formation, regenerates approximately one-third of the NH₂ to NH₃. About 20% of NH₂ reacts with HO₂ to produce H₂NO and HNO. The majority of H₂NO is converted to HNO through chemical reactions, and more than 95% of HNO reacts with O₂ and NO₂ to form NO. A small quantity of NH₂ reacts with NO_x to produce NHH and H₂NN, and H₂NN reacts with H, O, and NH₂ to produce NNH. The majority of NNH is converted to the final product N₂. The remaining NH₂ produces NO, N₂O, and N₂, and the produced NO and NO₂ continue to react with the remaining NH₂. Fig. 11 depicts the pathway analysis of oxidation primarily involving NO, NO₂, and N₂O to further examine the transformation between nitrogen-containing species.

Almost half of NO can be observed to generate NO₂ through the reaction NO + HO₂ = NO₂ + OH. Over 50% of NO will react with NH₂ to form NNH and N₂, while the remaining NO will produce N₂O. Approximately 60% of NO is produced by the reduction of NO₂, via HNO + NO₂ = HONO + NO and NH₂ + NO₂ = H₂NO + NO. For most NO₂, HONO is a more important substance due to their mutual conversion. Over 80% of N₂O is produced by NH₂ + NO₂ = N₂O + H₂O, and the majority of N₂O will produce N₂, indicating that NH₂ promotes the conversion of NO_x into N₂.

HCN is an explosive liquid and a marker for biomass burning emissions, which are produced by the combustion of nitrogen- and carbon-

containing fuels. Using the NGH-W mechanism to predict, the amount of HCN produced by the combustion of ammonia and methanol is minimal. The majority of HCN is derived from H₂CN and HCNH, and the only source of H₂CN and HCNH is CH₂NH. In contrast, during the co-combustion of ammonia and methanol, CH₃OH decomposes into CH₂OH, and CH₂OH further converts into CH₂O, after which CO₂ is ultimately produced. Only a small quantity of CH₂OH will generate CH₂NH, resulting in a minimum quantity of HCN.

5. Conclusions

Experiments were conducted on the oxidation of ammonia and methanol mixture in a jet-stirred reactor at 700–1200 K, with ammonia/methanol ratios of 2:3, 3:2, and 4:1, and equivalence ratios of = 0.8, 1.0, and 1.2. Species analysis of oxidation reactions in the JSR was conducted using the SVUV-PIMS method. After comparing experimental results with simulation results from the CEU-NH₃[26], Okafor[9], Glarborg[34], and Konnov[35] mechanisms, the Glarborg[34] mechanism was chosen as the foundation for the optimization of the ammonia/methanol combustion mechanism. The main conclusions can be summarised as the following:

Methanol accelerates ammonia's oxidation rate and increases its oxidation temperature. Nitrogen oxides (NO, NO₂, N₂O) become prevalent around 1000 K and decrease gradually as the temperature rises. Compared to existing mechanisms, the newly constructed mechanism accurately predicts the experimental results, despite some quantitative differences.

ROP and pathway analysis of the oxidation reactions of ammonia/methanol indicate that a high oxygen concentration is conducive to the aggregation of HO₂, which in turn promotes the formation of a large number of OH radicals and the oxidation of ammonia/methanol. CH₃OH and NH₃ are predominantly oxidized via CH₂O and NH₂.

The nitrogen-containing species in the oxidation reactions of ammonia/methanol were analyzed. NH₂ reacts with HO₂ and NO_x to produce a number of nitrogen-containing species. As the equivalence ratio increases, NO_x gradually decreases, and the majority of NO_x is converted to NO and N₂O, which then produce N₂. Due to the substantial conversion of CH₃OH to CH₂O, the amount of HCN produced in the reaction is negligible.

CRedit authorship contribution statement

Yuan Zhuang: Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Data curation,

Conceptualization. **Ruikang Wu**: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Xinyan Wang**: . **Rui Zhai**: Writing – review & editing, Resources, Project administration, Investigation. **Changyong Gao**: Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2023.129628>.

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